$\Delta H = \quad nC_P\,(T_2 - T_1)$

$\Delta H = \quad 10\ mol \times 75.3\ J\ mol^{-1}\ K^{-1} \times (373 - 298)\ K$

$\Delta H = \quad 56475\ J$

$\Delta H = \quad 56.475\ kJ$

3. **தீர்வு**

கொடுக்கப்பட்டவை

$$C_6H_6\,(l) + \frac{7}{2}\,O_2(g) \rightarrow 6\,CO_2\,(g) + 3\,H_2O\,(l)$$

ΔU at 25^0 C = –3268.12 kJ

T = 25^0 C = 298 K ;

$\Delta U = -3268.12\ kJ\ mol^{-1}$

ΔH = ?

$\Delta H = \Delta U + \Delta n_g RT$

$\Delta H = \Delta U + (n_p - n_r)\ RT$

$$\Delta H = -3268.12 + \left(6 - \frac{15}{2}\right) \times 8.314 \times 10^{-3}\ x\ 298$$

$\Delta H = \quad -3268.12$

$\qquad\quad - (1.5 \times 8.314 \times 10^{-3} \times 298)$

$\Delta H = \quad - 3268.12 - 3.72$

$\Delta H = \quad - 3271.84\ kJ\ mol^{-1}$

4. **தீர்வு**

கொடுக்கப்பட்டவை

$Mg(S) + Br_2(1) \rightarrow$

$MgBr_2(S)\ \Delta H_f^0 = -524\ KJ\ mol^{-1}$

பதங்கமாதல்

$Mg(S) \rightarrow Mg(g)$

$\Delta H_1^{\,0} = +148\ KJ\ mol^{-1}$

அயனியாதல்

$Mg(g) \rightarrow Mg^{2+}(g) + 2e^-$

$\Delta H_2^{\,0} = 2187\ KJ\ mol^{-1}$

ஆவியாதல்

$Br_2(1) \rightarrow Br_2(g)$

$\Delta H_3^{\,0} = +31\ KJ\ mol^{-1}$

பிரிகையடைதல்

$Br_2(g) \rightarrow 2Br(g)$

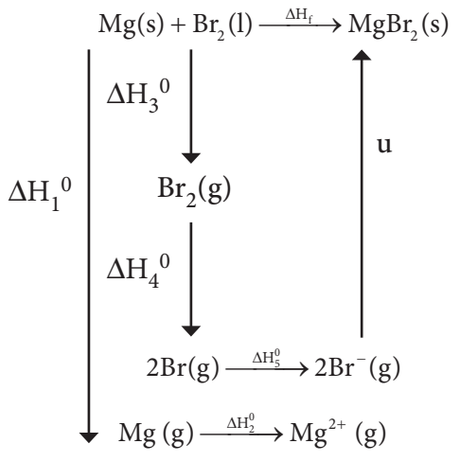
$\Delta H_4^{\,0} = +193\ KJ\ mol^{-1}$

எலக்ட்ரான் நாட்டம்

$Br(g) + e^- \rightarrow Br^-(g)$

$\Delta H_5^{\,0} = -\ 324.5\ KJ\ mol^{-1}$

தீர்வு



$\Delta H_f \quad = \quad \Delta H_1 + \Delta H_2 + \Delta H_3 + \Delta H_4$

$\qquad\qquad + 2\Delta H_5 + u$

$-\ 524 = \quad 148 + 2188 + 31 + 193$

$\qquad\qquad + (2 \times -\ 331) + u$

$-\ 524 = \quad 1897 + u$

$u = \quad -524 - 1911$

$u = \quad -\ 2435\ kJ\ mol^{-1}$

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